

Chapter 6. Model Neurons II: Conductances and Morphology

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Book: P. Dayan, L. Abbott: *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems (Computational Neuroscience Series)*, The MIT Press, 2015.

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The Cable Equation

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Single-compartment model

- V : the membrane potential,

$$c_m \frac{dV}{dt} = -i_m + \frac{I_e}{A}.$$

I_e : electrode current, A : membrane surface area of the cell, i_m : membrane current.

- All gating variables depending on V are determined by

$$\tau_z(V) \frac{dz}{dt} = z_\infty(V) - z, \quad z: \text{generic gating variable}.$$

$\tau_z(V)$, $z_\infty(V)$: experimental data.

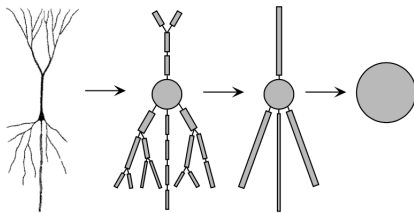


Figure 1: Figure 6.15 in the textbook

Connor-Stevens Model

- Hodgkin-Huxley model: in the single compartment model, consider the current i_e as the sum of a leakage current, a delayed-rectified K^+ current, and a transient Na^+ current:

$$i_m = \bar{g}_L(V - E_L) + \bar{g}_K n^4(V - E_K) + \bar{g}_{Na} m^3 h(V - E_{Na}).$$

- Connor-Stevens Model: with the same leak conductance $\bar{g}_L$ and reversal potential E_L , consider an extra K^+ conductance A-current:

$$i_m = \bar{g}_L(V - E_L) + \bar{g}_{Na} m^3 h(V - E_{Na}) + \bar{g}_K n^4(V - E_K) + \bar{g}_A a^3 b(V - E_A),$$

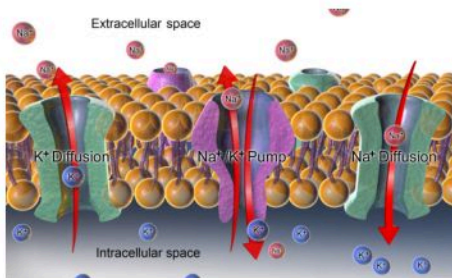


Figure 2: Resting potential. (2023, March 10). In Wikipedia.
https://en.wikipedia.org/wiki/Resting_potential

Connor-Stevens Model

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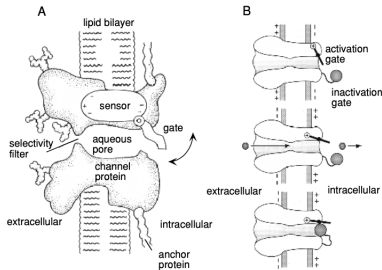


Figure 3: Figure 5.8 in the textbook

Connor-Stevens Model (continued)

- role of the additional A-current

Figure A: firing rate depends on the magnitude of the electrode current

Figure B: example of action-potential generation

Figure C: $\bar{g}_A = 0$

Figure D: A-current delays the occurrence of the first action potential.

- Neurons : type I (continuous firing rate from 0), type II (discontinuous)

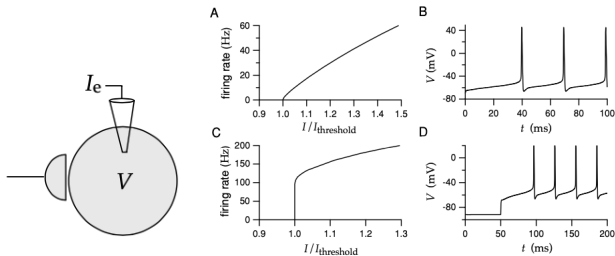


Figure 4: Figure 5.3 and 6.1 in the textbook

Postinhibitory Rebound and Bursting

- Huguenard and McCormick (1992): extend the Connor-Stevens model including a transient Ca^{2+} conductance:

$$i_{\text{CaT}} = \bar{g}_{\text{CaT}} M^2 H(V - E_{\text{Ca}}).$$

- A transient Ca^{2+} conductance acts like a slower version of Na^{+} conductance. A transient Ca^{2+} generates a slower transient depolarization called a Ca^{2+} spike. Generation of action potentials in response to release from hyperpolarization is called **postinhibitory rebound**.

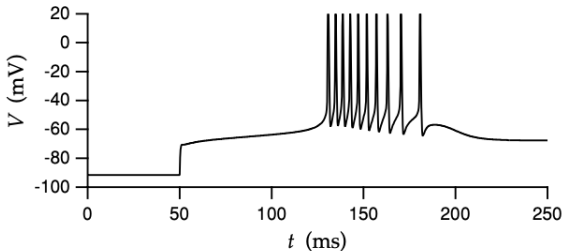


Figure 5: Figure 6.2 in the textbook

Postinhibitory Rebound and Bursting (continued)

- The transient Ca^{2+} current is an important component of models of thalamic relay neurons which exhibit different firing patterns in sleep and wakeful states.

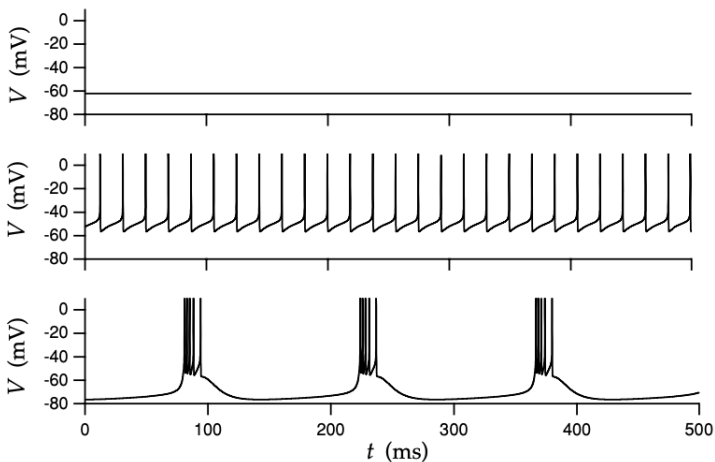


Figure 6: Figure 6.3 in the textbook : no electrode current, positive current(regular, periodic), negative current(periodic burst), in order.

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Cable theory

- Membrane potentials measured at different places within a neuron can take different values.

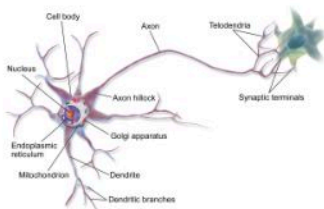


Figure 7: Axon. (2023, April 30). In Wikipedia.
<https://en.wikipedia.org/wiki/Axon>

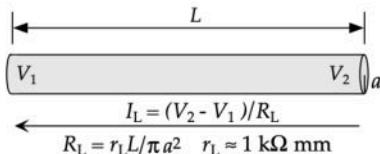


Figure 8: Figure 5.2 in the textbook ;
longitudinal current I_L , longitudinal resistance R_L , intracellular resistivity r_L

Cable theory

- Membrane potentials measured at different places within a neuron can take different values.
- **cable theory**: mathematical analysis of signal propagation within neurons.
- Membrane potential along a neuronal cable: $V(x, t)$ (x : longitudinal spatial coordinate)

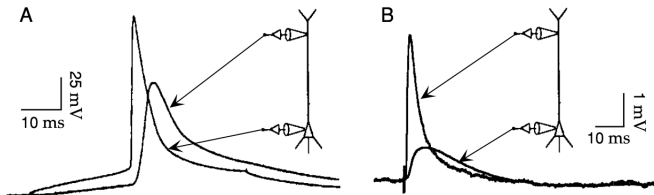


Figure 9: Figure 6.5 in the textbook ; delay and attenuation of an excitatory postsynaptic potential (EPSP) initiated in the dendrite by synaptic input as it spreads to the soma

Cable theory (Continued)

- Ohm's law ; $V(x + \Delta x) - V(x) =: \Delta V = -R_L I_L = -r_L \Delta x I_L / (\pi a^2)$.

$$I_L = -\frac{\pi a^2}{r_L} \lim_{\Delta x \rightarrow 0} \frac{\Delta V}{\Delta x} = -\frac{\pi a^2}{r_L} \frac{\partial V}{\partial x}.$$

Cable Equation

- Ohm's law ; $V(x + \Delta x) - V(x) = \Delta V = -R_L I_L = -r_L \Delta x I_L / (\pi a^2)$.

$$I_L = -\frac{\pi a^2}{r_L} \lim_{\Delta x \rightarrow 0} \frac{\Delta V}{\Delta x} = -\frac{\pi a^2}{r_L} \frac{\partial V}{\partial x}.$$

- Cable equation :

$$2\pi a \Delta x c_m \frac{\partial V}{\partial t} = - \left(\frac{\pi a^2}{r_L} \frac{\partial V}{\partial x} \Big|_{\text{left}} \right) + \left(\frac{\pi a^2}{r_L} \frac{\partial V}{\partial x} \Big|_{\text{right}} \right) - 2\pi a \Delta x (i_m - i_e).$$

$$\Rightarrow c_m \frac{\partial V}{\partial t} = \frac{1}{2ar_L} \frac{\partial}{\partial x} \left(a^2 \frac{\partial V}{\partial x} \right) - i_m + i_e.$$

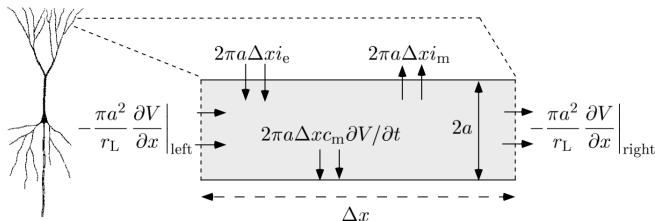


Figure 10: Figure 6.6 in the textbook.

Linear Cable Theory

- V_{rest} : the resting potential ; no net current flows across the membrane.

$$i_m = v/r_m, \quad v := V - V_{rest} \quad (\text{near the resting potential})$$

- If a is independent to x ,

$$c_m \frac{\partial v}{\partial t} = \frac{a}{2r_L} \frac{\partial^2 v}{\partial x^2} - \frac{v}{r_m} + i_e.$$

- $\tau_m := r_m c_m$, $\lambda := \sqrt{a r_m / (2 r_L)}$ (electrotonic length)

$$\tau_m \frac{\partial v}{\partial t} = \lambda^2 \frac{\partial^2 v}{\partial x^2} - v + r_m i_e.$$

- Two effects: form of cable equation and boundary condition (imposed at branching nodes and terminations).

Consider an infinite cable and a single branching node.

- Assumption: current is injected at a single point.
The solution for a point source of current injection is called a Green's function.

An Infinite Cable

- Assumption: the current being injected is held constant.

$$\lambda^2 \frac{d^2 v}{dx^2} = v - r_m i_e \quad \text{point source} \quad \xrightarrow{x \neq 0} \quad \lambda^2 \frac{d^2 v}{dx^2} = v \quad \Rightarrow \quad v(x) = B_1 e^{-\frac{x}{\lambda}} + B_2 e^{\frac{x}{\lambda}}.$$

- For an **infinite cable**, we require that the solution does not blow up when $x \rightarrow \pm\infty$:

$$B_1 = 0 \quad \text{for } x < 0, \quad B_2 = 0 \quad \text{for } x > 0.$$

- The continuity at $x = 0$ implies $B_1 = B_2 = B$; $v(x) = B e^{-|x|/\lambda}$.
- About B : study the Δx neighborhood A of $x = 0$ where the current is injected.
 - the current per unit area injected into the region A is $I_e/(2\pi a \Delta x)$.
 I_e : the total amount of current injected by the electrode.

$v'(x)$ is discontinuous at the point $x = 0$;

$$v'(-\Delta x/2) \approx B/\lambda, \quad v'(\Delta x/2) \approx -B/\lambda, \quad \Delta x \ll 1.$$

- Solve $\lambda^2(-2B/(\lambda \Delta x)) = -r_m I_e/(2\pi a \Delta x)$:

$$B = I_e R_\lambda / 2, \quad R_\lambda := \frac{r_m}{2\pi a \lambda}.$$

- The membrane potential for static current injection at the point $x = 0$ along an infinite cable is

$$v(x) = \frac{I_e R_\lambda}{2} \exp\left(-\frac{|x|}{\lambda}\right).$$

An Isolated Branching Node

- Situation: a single isolated junction of three semi-infinite cables.
 x : the coordinate, y : the distance away from the junction.

$$v_1(x) = p_1 I_e R_{\lambda_1} \exp(-x/\lambda_1 - y/\lambda_2),$$

$$v_2(x) = \frac{I_e R_{\lambda_2}}{2} [\exp(-|y - x|/\lambda_2) + (2p_2 - 1) \exp(-(y + x)/\lambda_2)],$$

$$v_3(x) = p_3 I_e R_{\lambda_3} \exp(-x/\lambda_3 - y/\lambda_2),$$

where, for $i = 1, 2$, and 3 ,

$$p_i = \frac{a_i^{3/2}}{a_1^{3/2} + a_2^{3/2} + a_3^{3/2}}, \quad \lambda_i = \sqrt{\frac{a_i r_m}{2r_L}}, \quad R_{\lambda_i} = \frac{r_L \lambda_i}{\pi a_i^2}.$$

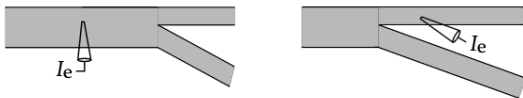


Figure 11: Figure 6.9 in the textbook.

The Rall Model

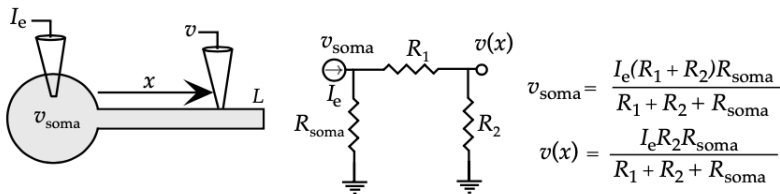


Figure 12: Figure 6.10 in the textbook.

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Multi-compartment Models

Multi-compartment Models

- When the complexities of real membrane conductances are included, the membrane potential is computed by splitting the neuron being modeled into separate regions or compartments.

Assumption: each compartment is small enough.

- Each compartment μ has its own membrane potential V_μ and the membrane current i_m^μ .

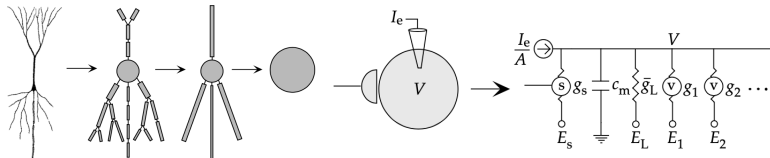


Figure 13: Figure 6.15 and 5.4 in the textbook ; a schematic diagram of a cortical pyramidal neuron and the equivalent circuit for a one-compartment neuron model

Multi-compartment Models (continued)

- Each compartment is coupled to two neighbors :

$$C_m \frac{dV_\mu}{dt} = -i_m^\mu + \frac{I_e^\mu}{A_\mu} + g_{\mu,\mu+1}(V_{\mu+1} - V_\mu) + g_{\mu,\mu-1}(V_{\mu-1} - V_\mu).$$

I_e^μ : the total electrode current flowing into compartment μ , A_μ : its surface area.

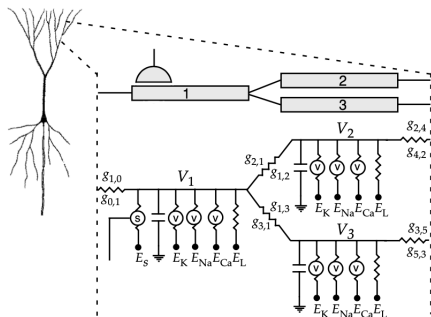


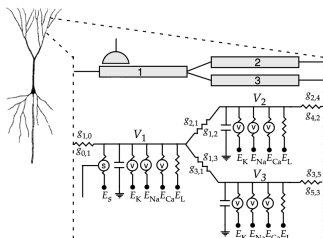
Figure 14: Figure 6.16 in the textbook ; a multi-compartment model of a neuron.

Multi-compartment Models (continued)

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$$C_m \frac{dV_\mu}{dt} = -i_m^\mu + \frac{I_e^\mu}{A_\mu} + g_{\mu,\mu+1}(V_{\mu+1} - V_\mu) + g_{\mu,\mu-1}(V_{\mu-1} - V_\mu).$$

I_e^μ : the total electrode current flowing into compartment μ , A_μ : its surface area.



Compartment μ (length L_μ , radius a_μ)
Compartment μ' (length $L_{\mu'}$, radius $a_{\mu'}$)

Resistance: $r_L L_\mu / (2\pi a_\mu^2) + r_L L_{\mu'} / (2\pi a_{\mu'}^2)$

$$\text{Inversion} \Rightarrow g_{\mu,\mu'} = \frac{a_\mu a_{\mu'}^2}{r_L L_\mu (L_\mu a_{\mu'}^2 + L_{\mu'} a_\mu^2)}.$$

Figure 15: Figure 6.16 in the textbook ; a multi-compartment model of a neuron.

Examples of Multi-compartment Modeling

- Action-Potential Propagation Along an (Un)myelinated Axon

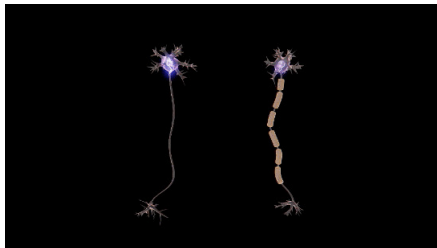


Figure 16: Myelin. (2023, March 10). In Wikipedia. <https://en.wikipedia.org/wiki/Myelin> ; Action potential propagation in myelinated neurons is faster than in unmyelinated neurons.

Action-Potential Propagation Along an Unmyelinated Axon

- The propagation velocity for an action potential is proportional to

$$\frac{\lambda}{\tau_m} = \sqrt{\frac{a}{2c_m^2 r_L r_m}}.$$

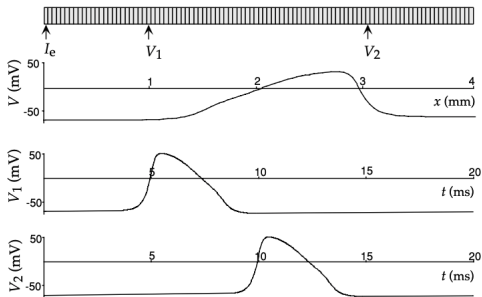


Figure 17: Figure 6.17 in the textbook ; Propagation of an action potential along a multi-compartment model axon ; 100 compartments.

Action-Potential Propagation Along an Myelinated Axon¹

- $\frac{\partial v}{\partial t} = D \frac{\partial^2 v}{\partial x^2}$, $D = \frac{a_1^2 \log(a_2/a_1)}{2c_m r_L d_m}$.
- Optimal choice $a_1 = a_2 e^{-1/2}$; $D = \frac{a_2^2}{4ec_m r_L d_m}$.

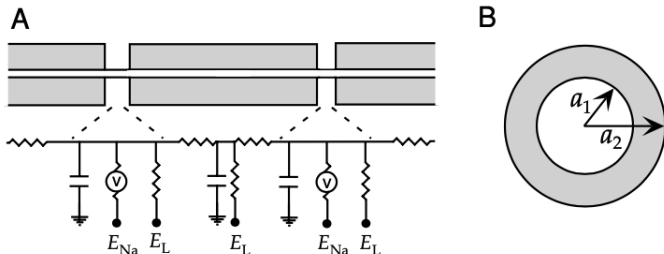


Figure 18: Figure 6.18 in the textbook.

¹Cf. R. Douglas Fields: *Myelin-More than Insulation*. Science **344** (2014) 264–266.

Thank you for your listening!